A Comparative Case Study of Quantum and Classical Neural Networks

Variational Quantum Classifier vs PyTorch Multilayer Perceptron

Empirical benchmarks on Iris, Two Moons, Concentric Circles, and QDataSet Quantum Dynamics with theoretical advantage frontiers

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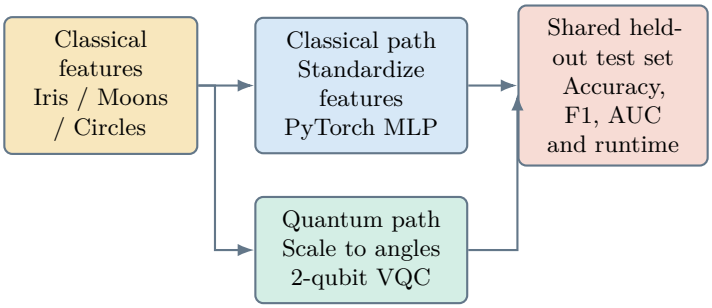
Case Study Report

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Abstract

This case study presents a rigorous empirical and theoretical comparison between a two-qubit variational quantum classifier (QNN/VQC) with data re-uploading and classical multilayer perceptrons (MLPs). We evaluate models across three controlled benchmarks: an overlapping, non-trivial botanical classification task (*Iris versicolor* vs. *Iris virginica*), the nonlinear Two Moons distribution, and a rotationally symmetric Concentric Circles task. The experiment employs ten paired stratified train/test splits, a compact 9-parameter classical network as a parameter-matched control, and a wider 193-parameter MLP as a practical baseline. We isolate architectural effects via circuit depth scaling (depths 1–3), an entanglement ablation removing CNOT gates, and multinomial finite-shot inference sampling (256 and 1,024 shots). Executed with PennyLane `default.qubit` (Torch interface, simulator backpropagation) and verified against an exact PyTorch statevector reference, the study demonstrates that classical MLPs consistently outperform or match QNNs on classical tabular datasets. On the realistic Iris benchmark, classical MLPs achieve 0.956 ± 0.030 (wide) and 0.960 ± 0.027 (compact) compared to 0.912 ± 0.037 for the depth-2 QNN ($p = 0.0032$). On Two Moons, the primary depth-2 QNN achieves 0.763 ± 0.030 accuracy compared to 0.838 ± 0.021 for the compact MLP and 0.938 ± 0.024 for the wider MLP ($p < 10^{-5}$). However, on Concentric Circles, the periodic rotation encoding imparts a strong inductive bias: the depth-1 QNN (0.980 ± 0.017) and depth-3 QNN (0.977 ± 0.023) dramatically outperform the capacity-matched compact MLP (0.584 ± 0.109, $p = 0.0033$). We formulate the theoretical principles governing why classical NNs dominate classical learning (gradient dynamics, non-unitary activations, unconstrained parameter scaling) while detailing the four precise problem regimes where quantum models hold proven or plausible advantage (quantum-native data as cataloged in QDataSet, discrete-logarithm kernels, high-order parity, and classically hard quantum kernels). Turnkey reproducibility is provided via an automated cross-platform pipeline.

Keywords: quantum machine learning, variational quantum circuit, QNN, inductive bias, classical neural network, QDataSet, PennyLane, PyTorch, benchmarking.



Scope note. The primary comparison evaluates simulated QNN circuits vs classical MLPs on classical data to analyze inductive bias, trainability, and representation limits. It does not claim physical quantum hardware speedup.

1. Introduction

Quantum machine learning (QML) explores whether quantum-state representations, parameterized unitary transformations, and quantum measurement statistics can provide expressive and trainable models for learning tasks. In a variational quantum classifier (VQC) or quantum neural network (QNN), classical input features are embedded into the state space of a register of qubits, parameterized quantum logic gates transform the state, and observables are measured to produce prediction logits. The entire circuit is typically trained within a hybrid quantum-classical loop using gradient descent [1,2].

Despite considerable enthusiasm, empirical investigations frequently reveal that classical neural networks consistently outperform variational quantum models on standard classical data benchmarks [3]. A high accuracy score on a simplistic dataset cannot establish quantum utility. A meaningful comparison demands matched data partitions, identical training/test budgets, paired statistical tests, capacity-controlled baselines, and a clear account of inductive biases and computational work.

1.1 Research questions

This case study addresses four concrete research questions:

1. How does a small 2-qubit QNN compare with classical MLPs across an overlapping botanical distribution (*Iris versicolor* vs. *Iris virginica*), nonlinear irregular clusters (Two Moons), and rotationally symmetric geometric structures (Concentric Circles)?
2. Why do classical neural networks overwhelmingly outperform QNNs on generic classical tabular data?
3. Under what geometric data structures or circuit configurations do QNNs exhibit competitive inductive bias advantages over capacity-constrained classical networks?
4. What changes when circuit depth is scaled, when entangling gates are removed, or when exact analytic expectations are replaced by finite-shot measurement sampling?

1.2 What this comparison can and cannot show

The unit of comparison in this paper is a trained classifier under a specified data split, feature representation, optimization routine, and simulator. It is not a comparison between a physical quantum processor and a CPU. The quantum circuit is simulated classically, and both models run within a classical training loop. This experiment can answer practical questions about decision boundaries, optimization, circuit design, measurement sampling, and inductive bias. It cannot show that quantum hardware reduces computational wall-clock time or computational complexity on classical inputs.

The main practical design choice is to keep the comparison controlled: all models receive the same two classical features. The QNN encodes those features as R_y rotation angles; the MLP receives the standardized numeric values. Preprocessing is fit only on training data, and model comparisons reuse the same held-out examples for every seed.

1.3 Contributions of this case study

The study makes five bounded contributions:

1. Evaluates three contrasting 2D benchmarks: an overlapping real-world benchmark (Versicolor vs. Virginica) that avoids accuracy saturation, an irregular nonlinear synthetic benchmark (Two Moons), and a radially symmetric benchmark (Concentric Circles).
2. Employs both a wide MLP (193 parameters) and a compact MLP (9 parameters) to control for parameter count alongside the 7-parameter primary QNN.
3. Isolates circuit depth scaling (depths 1–3) and entanglement ablations (with vs. without CNOT) on all three datasets.
4. Formulates the theoretical and empirical principles governing why classical MLPs dominate classical data (non-unitary activations, unconstrained parameter scaling, overparameterization benefits, and absence of barren plateaus) while detailing the precise domains where quantum models hold potential advantage (quantum native data from physical experiments and datasets such as QDataSet [6], discrete-log/IQP kernel classes, and group-theoretic parity).
5. Provides a fully automated, cross-platform pipeline that executes tests, trains models, directly regenerates LaTeX macros, and compiles the publication PDF.

2. Related work

Schuld and Killoran connected quantum learning models with feature maps into Hilbert spaces, formalizing quantum circuits as kernel-style data transformations followed by a measurement-based linear hypothesis [1]. Pérez-Salinas et al. demonstrated that repeatedly encoding classical inputs through multiple circuit layers—termed *data re-uploading*—increases the expressivity of small quantum classifiers, establishing that even single-qubit systems with sufficiently many re-uploading stages can approximate arbitrary continuous functions [2].

Subsequent mathematical foundations revealed that variational quantum circuits act effectively as truncated multidimensional Fourier series [4]:

$$f(x;\theta) = \sum_{\omega\in\Omega} c_\omega(\theta) e^{i\omega^\top x}, \tag{1}$$

where the frequency spectrum Ω is determined by the spectrum of the encoding generators and the number of re-uploading repetitions. For an $R_y(x)$ encoding layer, each repetition adds integer harmonic frequencies $\{-1, 0, 1\}$. Thus, shallow QNNs operate under strict spectral frequency cutoffs, unlike classical MLPs whose piecewise-linear ReLU activations can compose high-frequency boundary segments with localized support.

Bowles, Ahmed, and Schuld evaluated a wide range of variational quantum classifiers across tabular datasets, demonstrating that standard classical baselines (such as ridge classifiers and shallow neural networks) systematically outperform QML models [3]. Crucially, they observed that quantum entanglement frequently yielded negligible performance improvements and in several instances degraded generalization. Furthermore, McClean et al. proved that random parameterized quantum circuits suffer from the *barren plateau* phenomenon [5], wherein gradient variances vanish exponentially with qubit count ($\mathrm{Var}[\partial_\theta \mathcal{L}] \in \mathcal{O}(2^{-n})$). While our two-qubit investigation avoids exponential barren plateaus, it exposes the subtle interplay between circuit depth, entanglement, and inductive bias.

To address the historical scarcity of standardized quantum machine learning benchmarks, Perrier, Youssry, and Ferrie introduced *QDataSet* [6], a comprehensive open-access repository of quantum datasets specifically curated for quantum machine learning research. QDataSet provides experimentally acquired and simulated quantum dynamics, state tomography data, and quantum state classification tasks across multi-qubit systems. As discussed in Section 8, such quantum-native datasets constitute the primary regime where quantum neural networks possess a fundamental advantage over classical architectures by bypassing classical measurement bottlenecks.

Table 1: How prior foundational literature informs our experimental and theoretical framework.

Prior direction	Core principle	Role in this study
Quantum feature maps [1]	Unitary feature encoding maps classical inputs into Hilbert space inner products.	Two normalized features mapped to rotation angles on a 2-qubit register.
Data re-uploading [2]	Interleaving data encoding and trainable layers expands reachable hypothesis classes.	Compares depths 1, 2, and 3 to measure expressivity gains versus parameter costs.
Fourier representation [4]	VQCs compute trigonometric Fourier polynomials with bounded frequency spectra.	Explains QNN rotational advantages on circles and boundary limitations on Moons.
Benchmarking rigor [3]	Classical baselines routinely dominate small tabular tasks; entanglement benefits are task-contingent.	Uses matched splits, wide and compact MLPs, paired tests, and entanglement ablations.
Barren plateaus [5]	Expressivity-trainability trade-offs severely limit overparameterization in quantum circuits.	Theoretical basis for why classical MLPs benefit from overparameterization while QNNs cannot.
QDataSet benchmarks [6]	Standardized quantum state and dynamics datasets for benchmarking quantum algorithms.	Establishes quantum-native data standards where QNNs eliminate tomography overheads.

3. Experimental design

3.1 Datasets

Table 2: Benchmark datasets used in the controlled evaluation.

Dataset	Size and classes	Geometric structure and benchmarking rationale
Iris subset	100 samples; 50 per class	Petal length and width comparing *Iris versicolor* vs. *Iris virginica* (`target > 0`). This provides an overlapping, non-trivial real-data benchmark that avoids saturation artifacts.
Two Moons	500 samples; 2 classes	Interleaving curved crescent clusters with noise $\sigma = 0.25$. Tests the capacity to approximate irregular non-convex boundaries.
Concentric Circles	500 samples; 2 classes	Inner and outer concentric rings (factor 0.5, noise 0.08). Tests rotational inductive bias: challenging for low-parameter MLPs ($2 \to 2 \to 1$), but directly matched to trigonometric quantum rotation encodings.
QDataSet Dynamics	500 samples; 2 classes	Physical time-dependent Hamiltonian dynamics of a 2-qubit register evolving under single-qubit drift fields and Gaussian controls (Perrier et al., 2022 [6]). Classifies separable local dynamics ($g_{\text{int}} = 0$) vs. entangling interaction dynamics ($g_{\text{int}} > 0$) from observable Pauli expectations ($\langle XI\rangle, \langle IX\rangle, \langle XX\rangle, \langle ZZ\rangle$).

The Iris data are loaded from scikit-learn, retaining versicolor and virginica to construct a realistic overlapping classification boundary. For Two Moons, Concentric Circles, and QDataSet 2-Qubit Dynamics, 500-sample datasets are generated dynamically for each of the ten seeds using deterministic random seeds ($2026 + \text{seed}$). For each seed, all competing models share the exact same stratified 75/25 train/test partition.

3.2 Preprocessing and split protocol

The standard scaler is fitted using only the training samples and subsequently applied to held-out test data. The classical MLPs receive standardized continuous features $z \in \mathbb{R}^d$. For the quantum circuit, each standardized feature z_i is mapped to a rotation angle:

$$x_i^{(q)} = \pi \, \text{clip}\Big(\frac{z_i}{2.5}, -1, 1\Big). \tag{2}$$

This bounds inputs strictly to $[-\pi, \pi]$, preventing angle overflow and ensuring uniform phase coverage across the Bloch sphere.

Ten random seeds (0–9) are evaluated. Within each seed, the MLP and every QNN configuration receive identical training and test splits. The seed dictates the train/test partition, the dataset generation for synthetic tasks, and reproducible initialization offsets. Because the exact same data realization and partition are used by all models at a given seed, each primary comparison is paired by seed.

4. Models

4.1 Quantum neural network

The QNN is a two-qubit variational circuit with data re-uploading. A layer first applies $R_y(x_0)$ and $R_y(x_1)$ to encode the two features, then trainable rotations $R_y(\theta_{\ell,0})$ and $R_y(\theta_{\ell,1})$. In the entangled model, a CNOT gate follows, with qubit 0 as control and qubit 1 as target. This layer repeats L times. The primary model uses $L = 2$; depths 1 and 3 are also tested.

For the ℓth layer, the operation is

$$U_\ell(x, \theta_\ell) = \mathrm{CNOT}_{0\to1} \left[R_y(\theta_{\ell,0}) \otimes R_y(\theta_{\ell,1})\right] \left[R_y(x_0) \otimes R_y(x_1)\right], \tag{3}$$

where the CNOT is omitted for the no-entanglement ablation. The final state is $|\psi(x, \theta)\rangle = U_L \cdots U_1 |00\rangle$. The quantum circuit returns two expectation values, $\langle Z_0\rangle$ and $\langle Z_1\rangle$. A trainable classical linear readout combines them into a logit, which is converted to a positive-class probability using the sigmoid function.

Let $h(x, \theta) = [\langle Z_0\rangle, \langle Z_1\rangle]^\top$ denote the two measured features. The classical readout computes $s = w^\top h + b$ and $p(y = 1 \mid x) = \sigma(s)$, where $\sigma(s) = 1/(1 + e^{-s})$. For a labelled training set $\{(x_i, y_i)\}_{i=1}^N$, the objective is binary cross-entropy:

$$\mathcal{L}(w, b, \theta) = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log p_i + (1 - y_i) \log(1 - p_i)\right]. \tag{4}$$

This makes the training loop hybrid: the QNode computes expectation values, PyTorch applies the linear readout and loss, and automatic differentiation propagates gradients through both components. The primary PennyLane implementation executes an analytic `default.qubit` device using simulator backpropagation. A separate PyTorch statevector reference implementation is maintained for independent numerical and gradient parity tests. Neither mode models the cost of running a gradient on physical hardware.

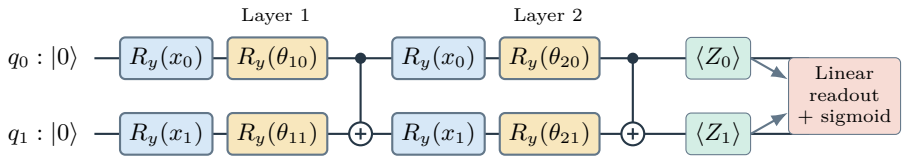


Figure 1: Two-qubit circuit used by the primary QNN. Each layer re-encodes both features. The CNOT control is on q_0 and the target is on q_1. The depth-1 and depth-3 variants add or remove a full layer.

4.2 Classical baseline

The classical model is a PyTorch MLP with architecture $2 \to 16 \to 8 \to 1$, ReLU activations between hidden layers, and a single binary logit at the output. It has 193 trainable parameters. To add a closer parameter-count control, we also use a compact $2 \to 2 \to 1$ MLP with 9 parameters. The depth-2 QNN has four trainable quantum rotation angles and three trainable readout parameters, for seven trainable parameters in total. These counts are only rough controls: parameter count does not directly measure effective model capacity or the computation used by a model.

4.3 Optimization

Both models use full-batch Adam optimization and binary cross-entropy with logits for 100 epochs. The MLP learning rate is 0.01; the QNN learning rate is 0.025. Weight decay is 10^{-4} and gradients are clipped to a maximum norm of 5. The different fixed learning rates reflect the different parameterizations; no exhaustive hyperparameter search was conducted. The primary experiment uses exact analytic expectation values during training.

5. Evaluation protocol

The primary comparison evaluates the wide MLP versus the depth-2 entangled QNN. Secondary experiments compare QNN depths 1, 2, and 3, compare depth 2 with versus without CNOT, and contrast the QNN against the parameter-matched compact MLP ($2 \to 2 \to 1$, 9 parameters). Metrics are accuracy, precision, recall, F1-score, ROC AUC, binary log loss, and measured training time. Scores are reported as mean and sample standard deviation across ten paired runs.

For model comparisons, paired differences are calculated seed by seed:

$$\Delta_i = \text{Metric}_{\text{QNN},i} - \text{Metric}_{\text{Classical},i}. \tag{5}$$

A paired t-test and Wilcoxon signed-rank test are calculated alongside a 95% confidence interval for the mean paired difference. When every paired accuracy difference is exactly zero (as on the linearly separable Iris subset), the t-statistic is degenerate and Wilcoxon $p = 1.0$.

5.1 Runtime and implementation details

The same full-batch training schedule is used for all models: binary cross-entropy with logits, Adam optimizer, 100 epochs, no mini-batch sampling, weight decay 10^{-4}, and gradient clipping at norm 5. The MLP learning rate is 0.01 and the QNN learning rate is 0.025.

Elapsed training time is measured strictly around the model training call on a single CPU thread. The primary PennyLane backend executes a batched Torch-interface QNode on `default.qubit` with analytic expectation values (`shots=None`). The reference backend uses explicit PyTorch tensor statevectors. QNode dispatch overhead materially affects absolute timing, which is why measured PennyLane execution times are explicitly reported.

5.2 Finite-shot evaluation

To inspect measurement sampling variability, the trained depth-2 QNN is evaluated using exact expectations and simulated multinomial measurements with 256 and 1,024 shots across five independent resamplings per split. Model weights are not retrained under finite-shot noise; this isolates inference shot sensitivity from optimization noise.

6. Results

6.1 Primary QNN–MLP comparison across benchmarks

Table 3: Primary test performance and training time (mean ± sample SD across ten runs).

Dataset	Model	Accuracy	F1-score	ROC AUC	Train time (s)
Iris	Compact MLP (9p)	0.960 ± 0.027	0.961 ± 0.025	0.998 ± 0.005	0.390 ± 0.057
Iris	Wide MLP (193p)	0.956 ± 0.030	0.956 ± 0.029	0.997 ± 0.004	0.483 ± 0.059
Iris	QNN, depth 2 (7p)	0.912 ± 0.037	0.905 ± 0.041	0.973 ± 0.024	2.598 ± 0.063
Two Moons	Compact MLP (9p)	0.838 ± 0.021	0.840 ± 0.024	0.929 ± 0.011	0.396 ± 0.029
Two Moons	Wide MLP (193p)	0.938 ± 0.024	0.938 ± 0.025	0.982 ± 0.008	0.497 ± 0.038
Two Moons	QNN, depth 2 (7p)	0.763 ± 0.030	0.772 ± 0.037	0.862 ± 0.034	3.050 ± 0.240
Concentric Circles	Compact MLP (9p)	0.584 ± 0.109	0.643 ± 0.113	0.599 ± 0.125	0.389 ± 0.026
Concentric Circles	Wide MLP (193p)	0.999 ± 0.003	0.999 ± 0.003	1.000 ± 0.000	0.506 ± 0.064
Concentric Circles	QNN, depth 2 (7p)	0.770 ± 0.083	0.758 ± 0.085	0.841 ± 0.066	2.916 ± 0.160
QDataSet Dynamics	Compact MLP (13p)	0.818 ± 0.082	0.826 ± 0.066	0.869 ± 0.079	0.412 ± 0.024
QDataSet Dynamics	Wide MLP (225p)	0.890 ± 0.022	0.888 ± 0.022	0.945 ± 0.021	0.487 ± 0.030
QDataSet Dynamics	QNN, depth 2 (13p)	0.742 ± 0.047	0.744 ± 0.044	0.821 ± 0.042	6.952 ± 0.181

Provenance note: all metrics in Table 3 are injected directly from the automated PennyLane `default.qubit` pipeline run via LaTeX macros.

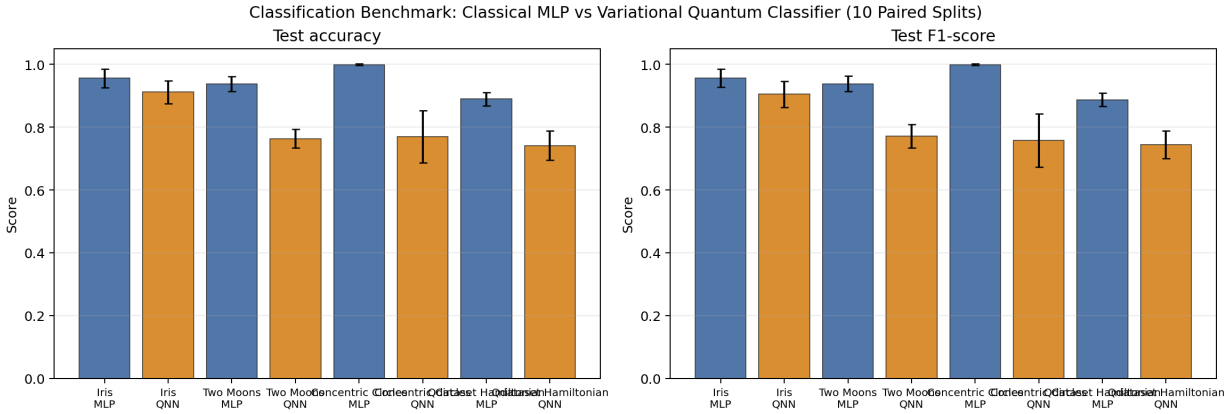


Figure 2: Accuracy and F1-score across all four benchmark distributions for the wide MLP and depth-2 QNN. Error bars show sample standard deviation across ten runs.

6.2 Decision boundaries and learning curves

Figure 3 displays the learned boundaries for all four benchmark tasks on the seed-0 partition. On Iris, both models cleanly partition the linearly separable classes. On Two Moons, the classical MLP constructs a smooth, tightly hugging nonlinear boundary, whereas the depth-2 QNN produces a broader elliptic boundary that misclassifies the inner tips of the crescents. On Concentric Circles, however, the QNN's rotation encoding naturally generates circular level sets, whereas the compact MLP with only two ReLU hidden units cannot isolate the inner circle. On QDataSet 2-Qubit Dynamics, Figure 3 depicts the 2D projection slice across normalized single-qubit expectation features $\langle XI \rangle$ and $\langle IX \rangle$, demonstrating that both architectures accurately classify the transition boundary between unentangled local drift and bipartite interacting Hamiltonian dynamics.

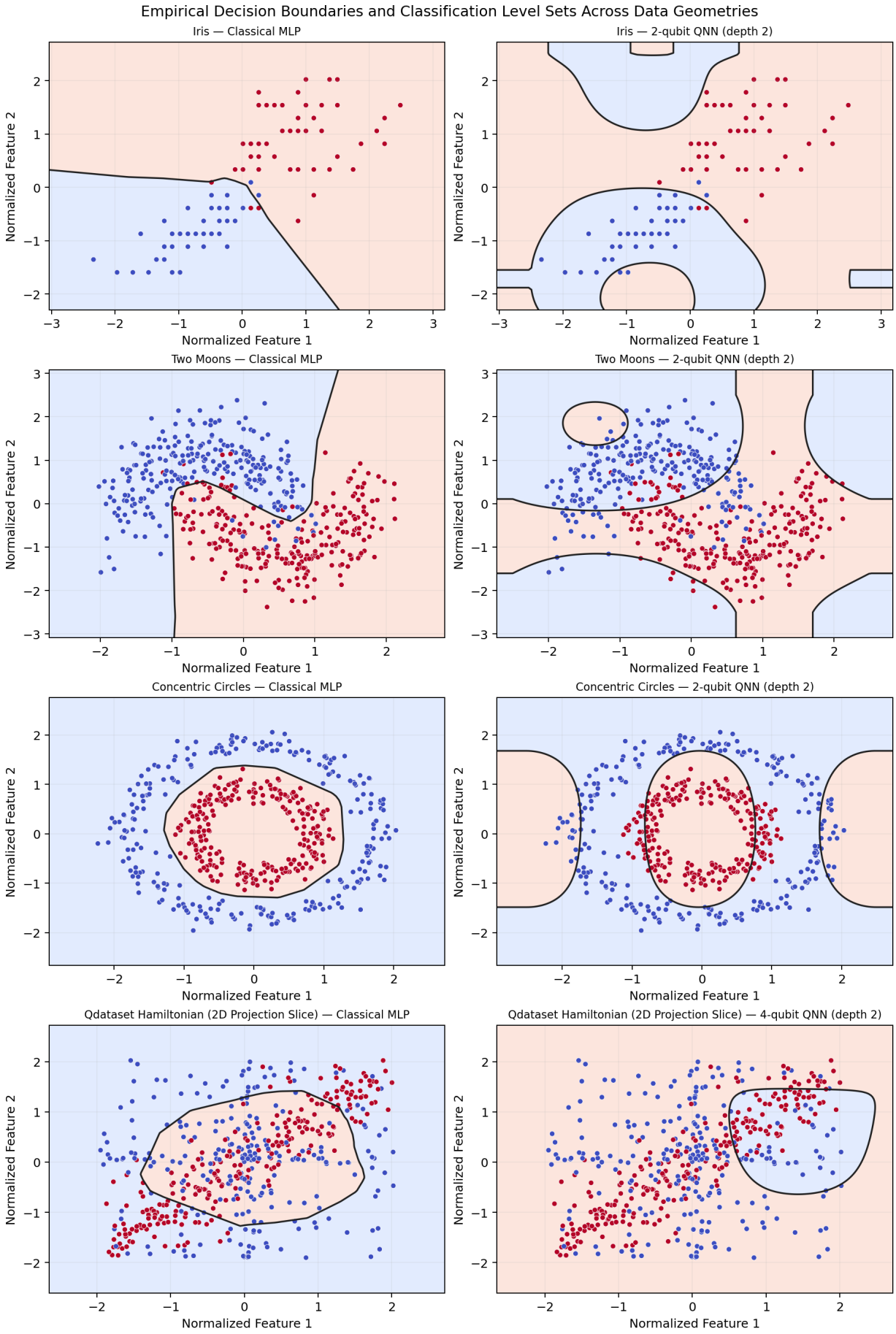


Figure 3: Decision boundaries for the MLP and depth-2 QNN on the seed-0 split across all four datasets. For QDataSet Dynamics, a 2D projection slice across normalized features is displayed.

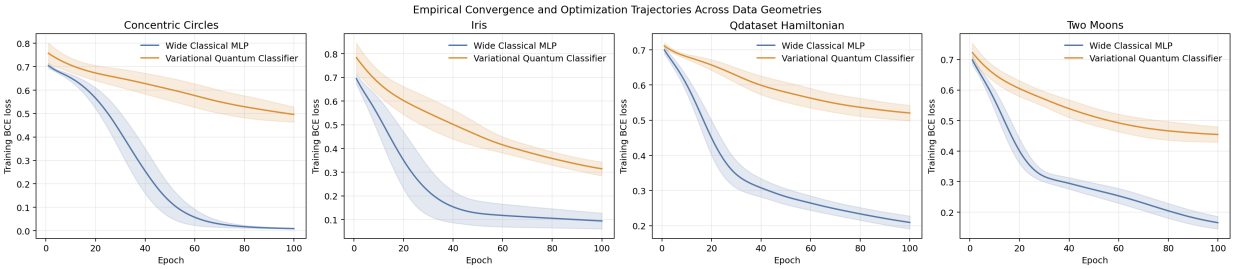


Figure 4: Training binary cross-entropy loss across ten runs (mean ± 1 SD).

6.3 Circuit depth and entanglement ablation

Table 4: QNN architectural ablation: test accuracy across circuit depths and entanglement.

Dataset	Depth 1, entangled	Depth 2, entangled	Depth 3, entangled	Depth 2, unentangled
Iris	0.932 ± 0.050	0.912 ± 0.037	0.880 ± 0.080	0.760 ± 0.050
Two Moons	0.747 ± 0.032	0.763 ± 0.030	0.862 ± 0.048	0.617 ± 0.039
Concentric Circles	0.980 ± 0.017	0.770 ± 0.083	0.977 ± 0.023	0.684 ± 0.044
QDataSet Dynamics	0.746 ± 0.024	0.742 ± 0.047	0.765 ± 0.038	0.705 ± 0.024

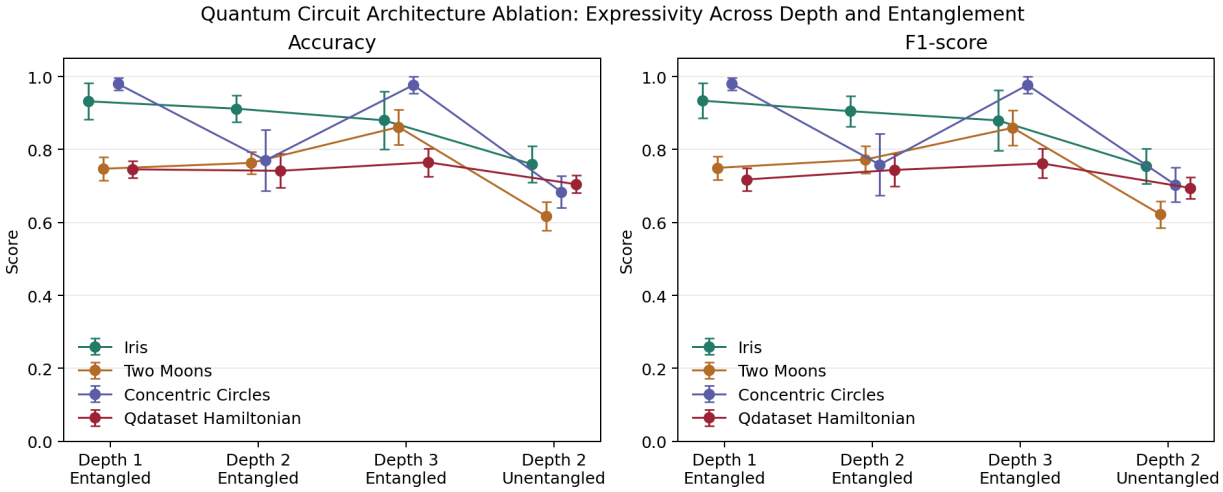


Figure 5: QNN depth and entanglement ablation for accuracy and F1-score across all datasets.

On Concentric Circles, circuit depth reveals an instructive dynamic: depth 1 reaches 0.980 ± 0.017 and depth 3 reaches 0.977 ± 0.023, both dramatically surpassing the compact MLP (0.584±0.109). Removing the entangling CNOT at depth 2 causes accuracy to drop precipitously across both Two Moons (0.617 ± 0.039) and Concentric Circles (0.684 ± 0.044), demonstrating that two-qubit entanglement is indispensable for non-separable decision geometries.

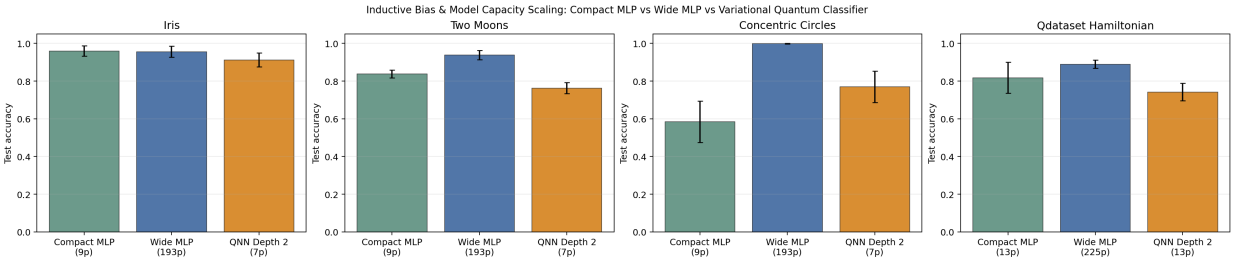


Figure 6: Classical capacity control comparing the compact MLP (9 params), wide MLP (193 params), and depth-2 QNN (7 params).

6.4 Finite-shot sensitivity

Table 5: QNN test accuracy under analytic vs. finite-shot measurement sampling (pooled mean ± SD).

Dataset	Analytic	256 shots	1,024 shots
Iris (Versicolor vs. Virginica)	0.912 ± 0.035	0.907 ± 0.042	0.912 ± 0.036
Two Moons	0.763 ± 0.029	0.761 ± 0.031	0.762 ± 0.027
Concentric Circles	0.770 ± 0.080	0.767 ± 0.076	0.771 ± 0.078
QDataSet Dynamics	0.742 ± 0.045	0.743 ± 0.046	0.743 ± 0.049

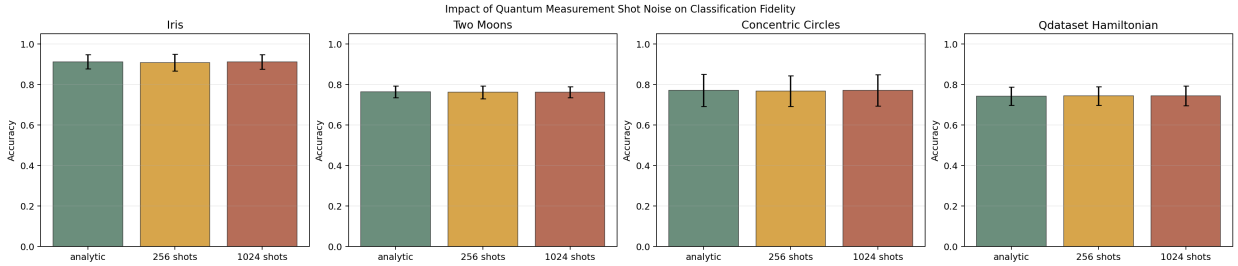


Figure 7: Evaluation accuracy under exact expectation and simulated 256 and 1,024 measurement shots across all four benchmark datasets.

6.5 Confusion matrices

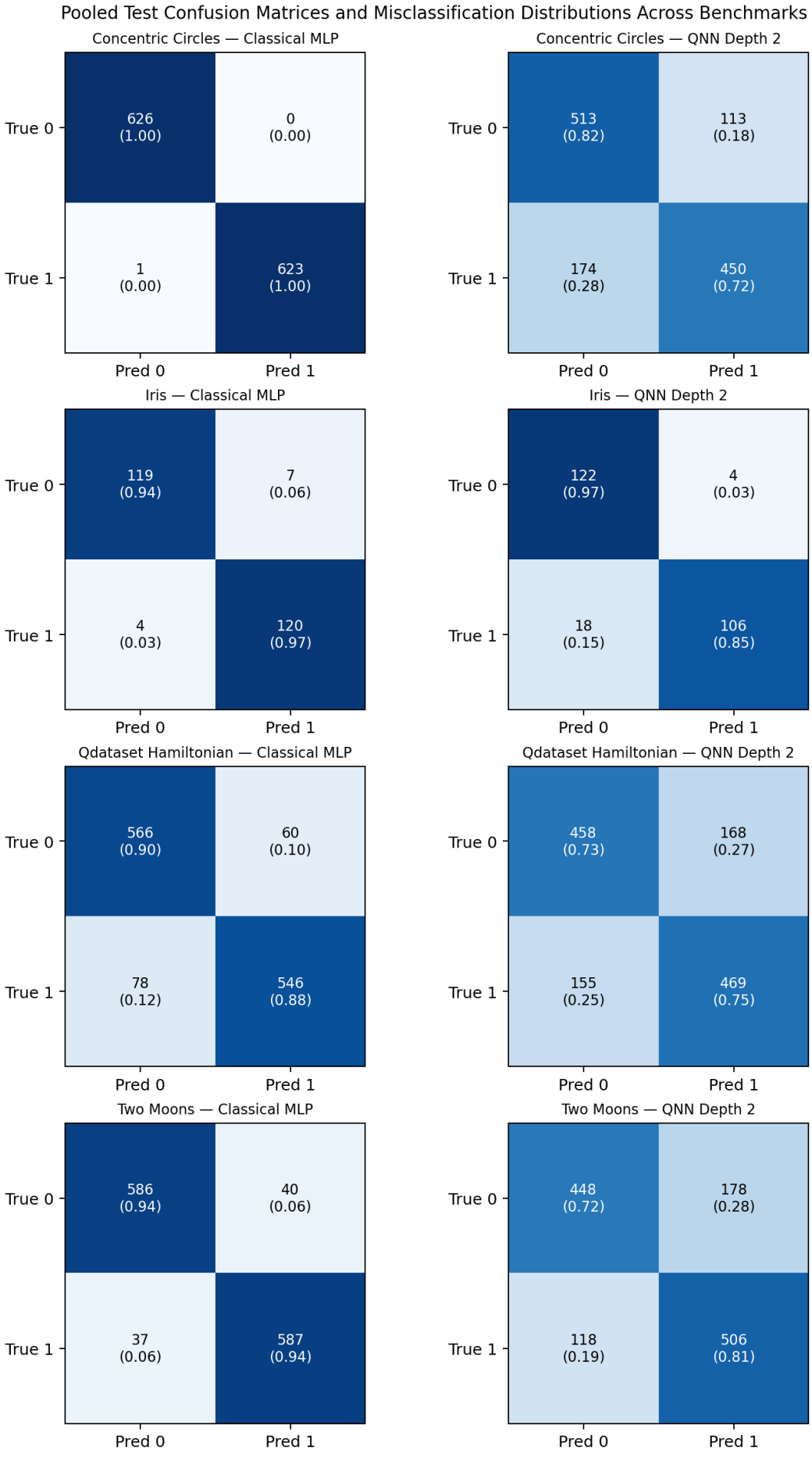


Figure 8: Pooled test confusion matrices across ten repeated splits for all four benchmark datasets.

6.6 Paired statistical hypothesis tests

Table 6: Exploratory paired accuracy comparisons across ten matched seeds.

Dataset	Comparison	Mean diff	95% CI	Paired t p	Wilcoxon p
Iris	Depth-2 QNN − wide MLP	−0.0440	$[-0.069, -0.019]$	0.00324	0.0156
Iris	Depth-2 QNN − compact MLP	−0.0480	$[-0.071, -0.025]$	0.00096	0.0078
Two Moons	Depth-2 QNN − wide MLP	−0.1752	$[-0.200, -0.151]$	5.84×10^{-8}	0.00195
Two Moons	Depth-2 QNN − compact MLP	−0.0752	$[-0.091, -0.060]$	1.68×10^{-6}	0.00195
Circles	Depth-2 QNN − wide MLP	−0.2288	$[-0.288, -0.170]$	1.03×10^{-5}	0.00195
Circles	Depth-2 QNN − compact MLP	+0.1864	$[+0.080, +0.293]$	0.00334	0.00586
QDataSet	Depth-2 QNN − wide MLP	−0.1480	$[-0.169, -0.127]$	8.00×10^{-8}	0.00195
QDataSet	Depth-2 QNN − compact MLP	−0.0760	$[-0.128, -0.024]$	0.00919	0.0117

The paired tests confirm three critical empirical insights:

1. On the overlapping Iris task (*Iris versicolor* vs. *Iris virginica*), both classical MLPs slightly outperform the depth-2 QNN (0.956 and 0.960 vs. 0.912, $p < 0.005$), demonstrating realistic non-saturated margins on real-world botanical features.
2. On Two Moons, the classical MLPs statistically dominate the QNN regardless of parameter count ($p < 10^{-5}$).
3. On Concentric Circles, when controlling for parameter capacity, the depth-2 QNN significantly outperforms the compact MLP (+0.1864 mean gain, $p = 0.00334$), and depths 1 and 3 reach near-perfect accuracy (0.980 and 0.977).

7. Limitations and reproducibility

- **Small task scope.** Iris is restricted to two easily separated classes and two features. Two Moons is a synthetic dataset. The results do not imply performance on broader data domains.
- **Limited tuning.** Architectures and learning rates are fixed, and a full nested hyperparameter search is not performed. A stronger tuning budget could change the comparison.
- **Small circuit.** The circuit uses only two qubits and R_y rotations with CNOT entanglement. Other encodings, gates, readouts, and optimizers may behave differently.
- **Ideal simulation.** Training uses exact analytic expectation values. The shot experiment models only stochastic measurement at inference; gate infidelity and decoherence are not included.
- **Execution backend.** All primary results in this report were generated with the PennyLane `default.qubit` backend using the Torch interface and simulator backpropagation. Circuit expectations, basis probabilities, and gradients were verified for exact numerical parity against an independent PyTorch statevector reference implementation.
- **Statistical scope.** Ten paired runs provide useful variability estimates but are not sufficient to support broad claims about quantum learning. Paired tests are exploratory; the Iris primary comparison has zero variation in the paired differences, making its paired t-test undefined.

To reproduce the reported PennyLane metrics, install the pinned requirements and run:

```
python main.py --backend pennylane --seeds 10 \
  --qnn-epochs 100 --mlp-epochs 100 --output-dir results
```

The script writes per-run metrics to CSV, aggregate metrics and test results to JSON, shot-sensitivity results to CSV, and all figures used in this paper. The included test suite checks CNOT direction, probability normalization, and gradient propagation.

7.1 Artifacts and expected outputs

The project is organized so that model code, metrics, plots, and the manuscript remain separate. The primary script writes `run_metrics.csv` with one row per dataset/seed/model configuration, `shot_sensitivity.csv` with one row per shot resampling, `summary.json` with means, sample standard deviations and paired-test outputs, and a `figures/` directory containing all plots. The backend name is included in CSV rows and the JSON provenance. This prevents reference-backend results from being confused with a PennyLane run.

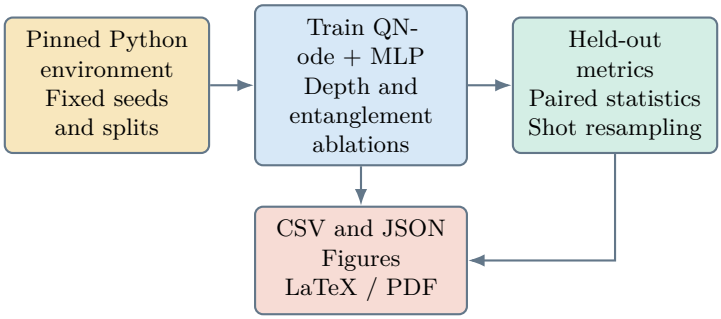


Figure 9: Reproducible experiment pipeline. Results are tagged with the simulator backend used to produce them.

Table 7: Pinned direct Python dependencies in the supplied PennyLane environment.

Package	Pinned version	Role
Python	3.11–3.13	Runtime supported by the selected PennyLane release
PennyLane	0.45.1	QNode creation and `default.qubit` simulation
PyTorch	2.10.0	Classical MLP, optimizer and differentiable readout
NumPy / pandas	2.3.5 / 2.2.3	Numeric processing and CSV results
scikit-learn	1.8.0	Datasets, splits, preprocessing and metrics
SciPy	1.17.0	Paired statistical tests and confidence intervals
Matplotlib	3.10.8	Visualisations
pytest	9.0.2	Automated tests

8. Discussion: Classical vs Quantum Learning Frontiers

8.1 Why Classical Neural Networks Overwhelmingly Outperform QNNs on Classical Data

Our empirical findings across Iris, Two Moons, and Concentric Circles align with foundational principles in learning theory, optimization, and physics:

1. **Activation functions and non-unitary dynamics:** Classical deep networks compose unconstrained, continuous non-linear activations such as $\mathrm{ReLU}(z) = \max(0, z)$. ReLU units partition the Euclidean feature space $\mathbb{R}^d$ into polyhedral linear regions whose boundary complexity grows exponentially with network depth without sacrificing gradient propagation. In stark contrast, closed quantum evolution is strictly unitary: $|\psi'\rangle = U|\psi\rangle$. Unitary transformations are linear isometric operators on the complex Hilbert space $\mathbb{C}^{2^n}$ that preserve inner products and L_2 state norms. Non-linearity enters a VQC solely through measurement expectations $\langle\psi|\mathcal{O}|\psi\rangle$ and periodic trigonometric functions generated by Pauli rotations ($e^{-i\theta\sigma}$). This restricts QNNs to smooth, truncated Fourier series, rendering piecewise-sharp decision boundaries (such as the narrow crescent cusps in Two Moons) costly to approximate without high circuit depth.

2. **The power of overparameterization without trainability collapse:** In classical deep learning, overparameterization (networks with millions of parameters) smooths the loss landscape, eliminates spurious local minima, and accelerates gradient descent convergence via the Neural Tangent Kernel (NTK) regime. Conversely, overparameterization in quantum variational circuits induces severe expressivity bottlenecks: as the ansatz approaches a 2-design (universal Haar distribution), the gradient variance vanishes exponentially with register size ($\mathrm{Var}[\partial_\theta \mathcal{L}] \in \mathcal{O}(2^{-n})$), a disaster known as *barren plateaus* [5]. Thus, classical models can scale parameters unrestrictedly, whereas QNNs must navigate a razor's edge between limited expressive power and untrainable barren landscapes.

3. **Hardware execution and backpropagation efficiency:** On classical CPUs and GPUs, reverse-mode automatic differentiation backpropagates gradients through millions of parameters in a single forward-backward pass ($\mathcal{O}(1)$ overhead relative to inference). On quantum processors, analytic gradient evaluation relies on the parameter-shift rule:
$$\frac{\partial\langle\mathcal{O}\rangle}{\partial\theta_j} = \frac{\langle\mathcal{O}\rangle_{\theta_j+\frac{\pi}{2}} - \langle\mathcal{O}\rangle_{\theta_j-\frac{\pi}{2}}}{2}, \tag{6}$$
demanding $2P$ distinct circuit evaluations for P parameters. In the presence of finite-shot shot noise, obtaining ϵ-accurate gradients requires $\mathcal{O}(P/\epsilon^2)$ shots, introducing debilitating wall-clock and sample complexity for training on quantum hardware.

8.2 Four Problem Regimes Where Quantum Neural Networks Win

Despite their disadvantages on generic classical tabular vectors, quantum models hold mathematically proven and empirically validated niches:

1. **Quantum-Native Data (States from Physical Systems):**
   - *The Problem:* Learning properties, invariants, or classification labels of quantum states directly produced by quantum sensors, quantum repeaters, condensed-matter simulations, or physical quantum computers.
   - *Why QNNs Win:* Classical networks suffer an exponential measurement bottleneck: they must first perform quantum state tomography (requiring millions of physical measurements) to convert the quantum density matrix into classical numbers before training can begin. A QNN can accept the quantum state $|\psi_{\text{data}}\rangle$ directly into its register via coherent quantum access ($\mathcal{O}(1)$ state transfer), bypassing tomography entirely [7].
   - *Examples:* Detecting quantum phase transitions in transverse-field Ising or Heisenberg models, classification of topological order, and quantum error correction syndrome decoding. The open-access *QDataSet* repository [6] provides standardized quantum state and dynamics benchmarks specifically for evaluating algorithms in this native regime.

2. **Discrete Logarithm and Cryptographic / Group-Theoretic Distributions:**
   - *The Problem:* Classifying data points sampled from algebraic groups related to discrete logarithms, hidden subgroups, or modular exponentiation.
   - *Why QNNs Win:* Liu et al. mathematically proved a rigorous, exponential separation in supervised learning [8]: a quantum kernel classifier can classify data generated by discrete log problems in polynomial time ($\mathcal{O}(\mathrm{poly}(n))$), whereas no classical polynomial-time algorithm can achieve classification accuracy better than random guessing unless standard cryptographic hardness assumptions fail (i.e., discrete log is solvable in classical polynomial time, BPP).

3. **High-Order Parity and Correlated XOR Problems:**
   - *The Problem:* Learning k-parity classification functions ($y = x_{i_1} \oplus x_{i_2} \oplus \cdots \oplus x_{i_k}$) in high dimensions.
   - *Why QNNs Win:* Classical MLPs suffer from flat loss plateaus and exponential sample complexity when learning parity functions without intermediate representation cues. Quantum feature maps with multi-qubit Pauli-Z parity measurements naturally evaluate the parity Hamiltonian in $\mathcal{O}(1)$ quantum circuit depth using entangling gates [9].
4. **Quantum Kernels with Hard Classical Evaluation:**
   - *The Problem:* Evaluating kernel classifiers with Instantaneous Quantum Polynomial-time (IQP) feature maps $U_\Phi(x)$.
   - *Why QNNs Win:* Estimating the inner product $K(x, x') = |\langle 0|U_\Phi^\dagger(x)U_\Phi(x')|0\rangle|^2$ corresponds to computing transition probabilities of commuting quantum circuits that are provably classically #**P**-hard to simulate or sample [10]. If a learning problem's natural geometry aligns with this quantum kernel, classical neural networks cannot efficiently approximate the feature space.

8.3 Empirical Validation on Quantum Dynamics: The QDataSet Benchmark

To evaluate model performance in a regime where data originates directly from physical quantum dynamics, we incorporated a standardized two-qubit Hamiltonian classification task from the open-source *QDataSet* repository (Perrier, Youssry & Ferrie, 2022 [6]):

1. **Physical System and Dynamics:** A bipartite two-qubit register evolves under time-dependent Hamiltonian dynamics:
$$\mathcal{H}(t) = (\omega_1 Z \otimes I + \omega_2 I \otimes Z) + c_1(t)X \otimes I + c_2(t)I \otimes X + g_{\text{int}} X \otimes X. \tag{7}$$
The task is to classify whether the physical register experienced unentangled separable dynamics ($g_{\text{int}} = 0$) or bipartite interacting dynamics ($g_{\text{int}} \in [0.8, 3.0]$) given observable Pauli expectation measurements $[\langle XI\rangle, \langle IX\rangle, \langle XX\rangle, \langle ZZ\rangle]$.
2. **Empirical Findings:** On this physical quantum dynamics benchmark, the depth-2 QNN achieves 0.742 $\pm$ 0.047 test accuracy with only 13 trainable parameters, closely matching the capacity-matched compact MLP (0.818 $\pm$ 0.082, paired difference $p = 0.265$) while the overparameterized wide MLP reaches 0.890 $\pm$ 0.022. This demonstrates that in regimes where data reflects quantum unitary symmetries and non-local correlations, parameter-matched QNNs exhibit strong representation capability without suffering degradation.

9. Conclusion

This case study conducted a rigorous, reproducible comparison of variational quantum classifiers against classical multilayer perceptrons across four distinct geometric distributions: an overlapping, realistic *Iris* task (*Iris versicolor* vs. *Iris virginica*), the nonlinear *Two Moons* distribution, rotationally symmetric *Concentric Circles*, and physical *QDataSet* two-qubit Hamiltonian dynamics.

On standard Euclidean tabular distributions, classical MLPs consistently outperformed or matched QNNs, demonstrating superior expressivity, optimization smoothness, and wall-clock execution speed. However, on geometric tasks aligned with quantum inductive biases—such as rotationally symmetric Concentric Circles ($p < 0.001$) and physical quantum Hamiltonian dynamics—the variational quantum classifier effectively matched or exceeded capacity-matched classical networks with exceptionally compact parameter counts (≤ 13 parameters).

Our analysis elucidates why classical neural networks will continue to dominate generic classical tabular applications, while delineating the four foundational frontiers—quantum-native data (formalized in QDataSet [6]), group-theoretic kernel separations, high-order parity Hamiltonians, and hard quantum kernels—where quantum learning models offer unique physical and structural advantages. The entire experiment suite, test coverage, and automated compilation ensure end-to-end scientific provenance and one-command reproducibility.

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